# ELECTRÓNICA DIGITAL 3

## CLASE 7

## Módulo DAC

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# Módulo DAC: Configuración

La LPC1769 cuenta con un solo bloque conversor digital-analógico en escalera de resistencias conformando un divisor de tensión de Kelvin-Varley, el cual presenta una tasa de conversión de **10-bit** a una frecuencia máxima de conversión $f_{DAC_{max}}$ de 1 MHz. El valor digital $VALUE_D$ de 10-bit que se convertirá a analógico $VALUE_A$ se almacena en los bits VALUE[15:6].

**D/A Converter Register (DACR - address 0x4008 C000) bit description**

| Bit | Symbol | Value | Description | Reset Value |
|---|---|---|---|---|
| 15:6 | VALUE | | After the selected settling time after this field is written with a new VALUE, the voltage on the AOUT pin (with respect to $V_{SSA}$) is VALUE × (($V_{REFP}$ - $V_{REFN}$)/1024) + $V_{REFN}$. | 0 |

Previo a esta conversión se establece el tiempo de establecimiento $t_s$ de la misma, dado por el bit BIAS[16]:

$$BIAS = 0 \Longrightarrow t_{s(max)} = 1\mu s \longrightarrow \begin{cases} I_{max} = 700\mu A \\ f_{DAC_{max}} = 1\text{MHz} \end{cases} \Longrightarrow Rapidez \uparrow \; - Consumo \; \uparrow\downarrow$$

$$BIAS = 1 \Longrightarrow t_{s(max)} = 2{,}5\mu s \longrightarrow \begin{cases} I_{max} = 350\mu A \\ f_{DAC_{max}} = 400\text{KHz} \end{cases} \Longrightarrow Rapidez \downarrow \; - Consumo \; \downarrow$$

**D/A Converter Register (DACR - address 0x4008 C000) bit description**

| Bit | Symbol | Value | Description | Reset Value |
|---|---|---|---|---|
| 16 | BIAS[1] | 0 | The settling time of the DAC is 1 μs max, and the maximum current is 700 μA. This allows a maximum update rate of 1 MHz. | 0 |
| | | 1 | The settling time of the DAC is 2.5 μs and the maximum current is 350 μA. This allows a maximum update rate of 400 kHz. | |

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***Nota:*** *Estos tiempos son válidos para cargas cuya capacitancia en el pin AOUT no exceda los* $100\ pF$*. Un valor superior de impedancia, causara tiempos de establecimientos mayores a los especificados.*

> [1] The settling times noted in the description of the BIAS bit are valid for a capacitance load on the AOUT pin not exceeding 100 pF. A load impedance value greater than that value will cause settling time longer than the specified time. One or more graph(s) of load impedance vs. settling time will be included in the final data sheet.

Luego de este tiempo de estabilización seleccionado, la tensión en AOUT se determina como:

$$VALUE_A = VALUE_D \frac{(V_{REFP} - V_{REFN})}{2^{10}} + V_{REFN}$$

Sean $V_{REFP} = V_{DD} = 3{,}3V$ y $V_{REFN} = V_{SS} = 0V$, resulta:

$$\boxed{VALUE_A = VALUE_D \frac{3{,}3V}{1024}}$$

**D/A Pin Description**

| Pin | Type | Description |
|---|---|---|
| AOUT | Output | **Analog Output.** After the selected settling time after the DACR is written with a new value, the voltage on this pin (with respect to $V_{SSA}$) is VALUE × (($V_{REFP}$ - $V_{REFN}$)/1024) + $V_{REFN}$.<br>**Remark:** The DAC output is disabled when the device is in deep-sleep, power-down, or deep power-down mode. |